

We construct a rooted tree T whose nodes are the elements of $\mathcal{C}$ using the following standard recursive definition. The root of T is some arbitrarily chosen element of $\mathcal{C}$. The *children* of a node C are all the elements of $\mathcal{C}$ that are reachable from C via a single bridge, other than the parent of C in T (the root of T is the only node without a parent). The leaves of T are the nodes without children. Notice that since $\mathcal{C}$ contains at least two elements, so does T , and hence the root is not a leaf. For convenience, we order the children of a node C from left to right and denote by $\ell(C)$ the leftmost child and by $r(C)$ the rightmost child. It may be that $\ell(C) = r(C)$ if C has only one child, and for leaves we define $\ell(C) = r(C) = \emptyset$. Finally, we define $L(C)$ to be the leftmost leaf in the subtree of T rooted at C (if C is a leaf then $L(C) = C$).

Let $\mathcal{L} \subset \mathcal{C}$ denote the set of leaves of T . Notice that $\mathcal{L}$ is, in fact, a subset of $\mathcal{D}$ because each singleton in $\mathcal{S}$ is incident with at least three bridges, while the leaves of T are incident with only one bridge; the bridge connecting them to their parent in T . The same reasoning shows that for $s \in \mathcal{S}$, we have that s has at least two children in T and hence $\ell(s) \neq r(s)$. It now follows that $|\mathcal{S}| \leq |\mathcal{L}|$. Indeed, the mapping $s \rightarrow L(r(s))$ is one-to-one from $\mathcal{S}$ to $\mathcal{L}$.

Another important feature of the elements of $\mathcal{L}$ is that they each contain at least four vertices. Indeed, if $X \in \mathcal{L}$ then X is a connected bridgeless subgraph of G that is incident with only one bridge. Any vertex of X other than the one incident with that bridge has all of its neighbors in X , and as G has minimum degree at least 3, we have $|X| \geq 4$. We further partition $\mathcal{S}$ into two parts, $\mathcal{S}'$ are all those singletons s for which $L(r(s))$ has cardinality 4, and $\mathcal{S}''$ are all those singletons s for which $L(r(s))$ has cardinality at least 5. We now have:

$$|\mathcal{S}| + 4|\mathcal{S}'| + 5|\mathcal{S}''| = 5|\mathcal{S}'| + 6|\mathcal{S}''| \leq n. \quad (1)$$

We now color E making G rainbow connected. The edges of each element $X \in \mathcal{D}$ are colored using $4|X|/5 - 1$ dedicated colors. This can be achieved using Proposition 2.4. We also color the unique bridge connecting X to its parent in T (unless X is the root of T) using one dedicated color, altogether using $4|X|/5$ colors. Likewise, for each $s \in \mathcal{S}$ we color the unique bridge connecting s to its parent in T (unless s is the root of T) using one dedicated color. Notice that this process colors *all* the edges of G . Altogether we have used at most

$$|\mathcal{S}| - 1 + \frac{4}{5} \sum_{X \in \mathcal{D}} |X| = \frac{4}{5}n + \frac{1}{5}|\mathcal{S}| - 1$$

colors. There is, however, some spare in the estimate above. Recall that there are at least $|\mathcal{S}'|$ elements of $\mathcal{D}$ that have cardinality 4. Since 2-connected graphs with cardinality $|X| = 4$ can be rainbow connected using only two colors, we actually *do not* use $4|X|/5 - 1$ colors in this case; rather, we use only $4|X|/5 - 6/5$ colors, sparing an additional $1/5$ in the above calculation at least $|\mathcal{S}'|$ times. It follows that

$$rc(G) \leq \frac{4}{5}n + \frac{1}{5}|\mathcal{S}''| - 1.$$

By (1), $|\mathcal{S}''| \leq n/6$. It follows that $rc(G) < 5n/6$, as claimed. ■

2.2 General minimum degree

Proof of Theorem 1.4: A set of vertices S of G is called a *connected 2-dominating set* if S induces a connected subgraph of G , and, furthermore, each vertex outside of S has at least two neighbors in S . Let $\gamma_{2c}(G)$ denote the smallest size of a connected 2-dominating set of G . Notice that the parameter is well-defined since, trivially, $V(G)$ is a connected 2-dominating set of G ; hence $\gamma_{2c}(G) \leq n$.

We claim that $rc(G) \leq \gamma_{2c}(G) + 1$. Indeed, let S be a connected 2-dominating set with $|S| = \gamma_{2c}(G)$. As S induces a connected subgraph, we have $rc(S) \leq \gamma_{2c}(G) - 1$. Using two additional colors, say, red and blue, we can color the rest of the graph. For a vertex $v \in V \setminus S$, let (v, x) and (v, y) be two distinct edges with $x, y \in S$. We color (v, x) red and (v, y) blue, and do the same for each $v \in V \setminus S$. The resulting coloring clearly makes G rainbow connected.

A special case of the main result in [5], which was also implicitly proved earlier in [4] asserts that $\gamma_{2c}(G) \leq n \frac{\ln \delta}{\delta} (1 + o_\delta(1))$. Together with the argument in the previous paragraph we have that $rc(G) \leq n \frac{\ln \delta}{\delta} (1 + o_\delta(1))$, as required.

For the second, non-asymptotic part of the theorem, consider the following probabilistic argument (see, e.g. [1]). We create a subset of vertices X by randomly and independently choosing each vertex to X with probability p . Hence, the expected number of elements of X is $E[|X|] = np$. Let Y be the set of vertices not belonging to X and having no neighbor in X . The probability that $v \in Y$ is at most $(1 - p)^{\delta+1}$. Hence, $E[|Y|] \leq n(1 - p)^{\delta+1}$. Let Z be the set of vertices not belonging to X and having precisely one neighbor in X . The probability that $v \in Z$ is at most $\delta p(1 - p)^\delta$. Hence, $E[|Z|] \leq n\delta p(1 - p)^\delta$. Notice that $W = X \cup Y \cup Z$ is a 2-dominating set, although not necessarily a connected one. However, notice that the number of connected components in W is at most $|X| + |Y|$ (vertices of Z already have neighbors in X so they do not contribute additional connected components). Now, to make W connected, notice that it suffices to add just $2(|X| + |Y| - 1)$ additional vertices to W , forming a connected 2-dominating set S of G . The expected size of S is, henceforth,

$$E[|S|] \leq 3np + 3n(1 - p)^{\delta+1} + n\delta p(1 - p)^\delta - 2.$$

Since $\gamma_{2c}(G) \leq E[|S|]$ we have that

$$rc(G) \leq 3np + 3n(1 - p)^{\delta+1} + n\delta p(1 - p)^\delta.$$

Choosing $p = \ln \delta / \delta$ we get that

$$rc(G) < \frac{n}{\delta} (4 \ln \delta + 3).$$

■

As noted in the introduction, Theorem 1.4 cannot be improved below $\frac{3n}{\delta+1} - \frac{\delta+7}{\delta+1}$. To see this, we construct a connected n -vertex graph with minimum degree δ and diameter $\frac{3n}{\delta+1} - \frac{\delta+7}{\delta+1}$. Take m copies of $K_{\delta+1}$, denoted $X_1, \dots, X_m$ and label the vertices of X_i

with $x_{i,1}, \dots, x_{i,\delta+1}$. Take two copies of $K_{\delta+2}$, denoted X_0, X_{m+1} and similarly label their vertices. Now, connect $x_{i,2}$ with $x_{i+1,1}$ for $i = 0, \dots, m$ with an edge, and delete the edges $(x_{i,1}, x_{i,2})$ for $i = 0, \dots, m+1$. The obtained graph has $n = (m+2)(\delta+1) + 2$ vertices, and minimum degree δ (and maximum degree $\delta+1$). It is straightforward to verify that a shortest path from $x_{0,1}$ to $x_{m+1,2}$ has length $3m+5 = \frac{3n}{\delta+1} - \frac{\delta+7}{\delta+1}$. If δ is odd, we can, in fact, make the graph regular. Just delete a maximum matching within X_0 where the only non-matched vertex is $x_{0,1}$ and delete a maximum matching within X_{m+1} where the only non-matched vertex is $x_{m+1,2}$.

2.3 Regular graphs

Proof of Theorem 1.3: We are given a connected graph G with n vertices, and a set of pairwise vertex-disjoint cycles $C_1, \dots, C_t$ that cover $n - s$ vertices. Let c_i denote the length of C_i for $i = 1, \dots, t$. As in Lemma 2.1 we notice that by adding to the set of cycles an additional set of $s + t - 1$ edges we obtain a connected *spanning* subgraph of G that contains t cycles and $s + t - 1$ bridges. We color the bridges using $s + t - 1$ dedicated colors. Also, each even cycle of length c_i is colored using $c_i/2$ dedicated colors and each C_i which is a triangle is colored using one dedicated color. Now, let us arbitrarily pair the odd cycles of length at least 5. Without loss of generality assume that $C_1, \dots, C_p$ are the odd cycles of length at least 5. We pair C_{2i-1} with C_{2i} for $i = 1, \dots, \lfloor p/2 \rfloor$ (if p is odd then C_p remains unpaired, and we color it using $(c_p + 1)/2$ dedicated colors). We will color these pairs of odd cycles using the following procedure. While there remains at least one pair that is yet uncolored, pick a pair (C_{2i-1}, C_{2i}) with the property that there is an already colored path between a vertex x of C_{2i-1} and a vertex y of C_{2i} (there must be at least one such pair). Let e be the unique edge of C_{2i-1} opposite to x (in an odd cycle each vertex has a unique edge opposite to it). Let f be the unique edge of C_{2i} opposite to y . Assign to e and f the same dedicated color, assign $(c_{2i-1} - 1)/2$ dedicated colors to properly color the remaining $c_{2i-1} - 1$ edges of C_{2i-1} , and assign $(c_{2i} - 1)/2$ dedicated colors to properly color the remaining $c_{2i} - 1$ edges of C_{2i} . Notice that we used $(c_{2i-1} + c_{2i})/2$ new colors to color this pair, and that after this coloring every connected component of the subgraph of colored edges is properly rainbow connected.

Altogether, if we let ℓ denote the number of triangles in our set of cycles we have that the number of colors used to rainbow-connect G is

$$\ell + \sum_{i=1}^{t-\ell} c_i/2 + s + t - 1/2 = \ell + \frac{1}{2}(n - s - 3\ell) + s + t - 1/2.$$

(If p is even then we even have 1 instead of $1/2$ in both sides of the last equality.) Now, since $t \leq \ell + (n - s - 3\ell)/4$ we have, together with the last equality, that

$$rc(G) < 3n/4 + s/4 - 1/2$$

and the result follows. For the specific parts of the theorem notice that the first case follows from the fact that in a graph with a 2-factor we can assume $s = 0$ in the last inequality.